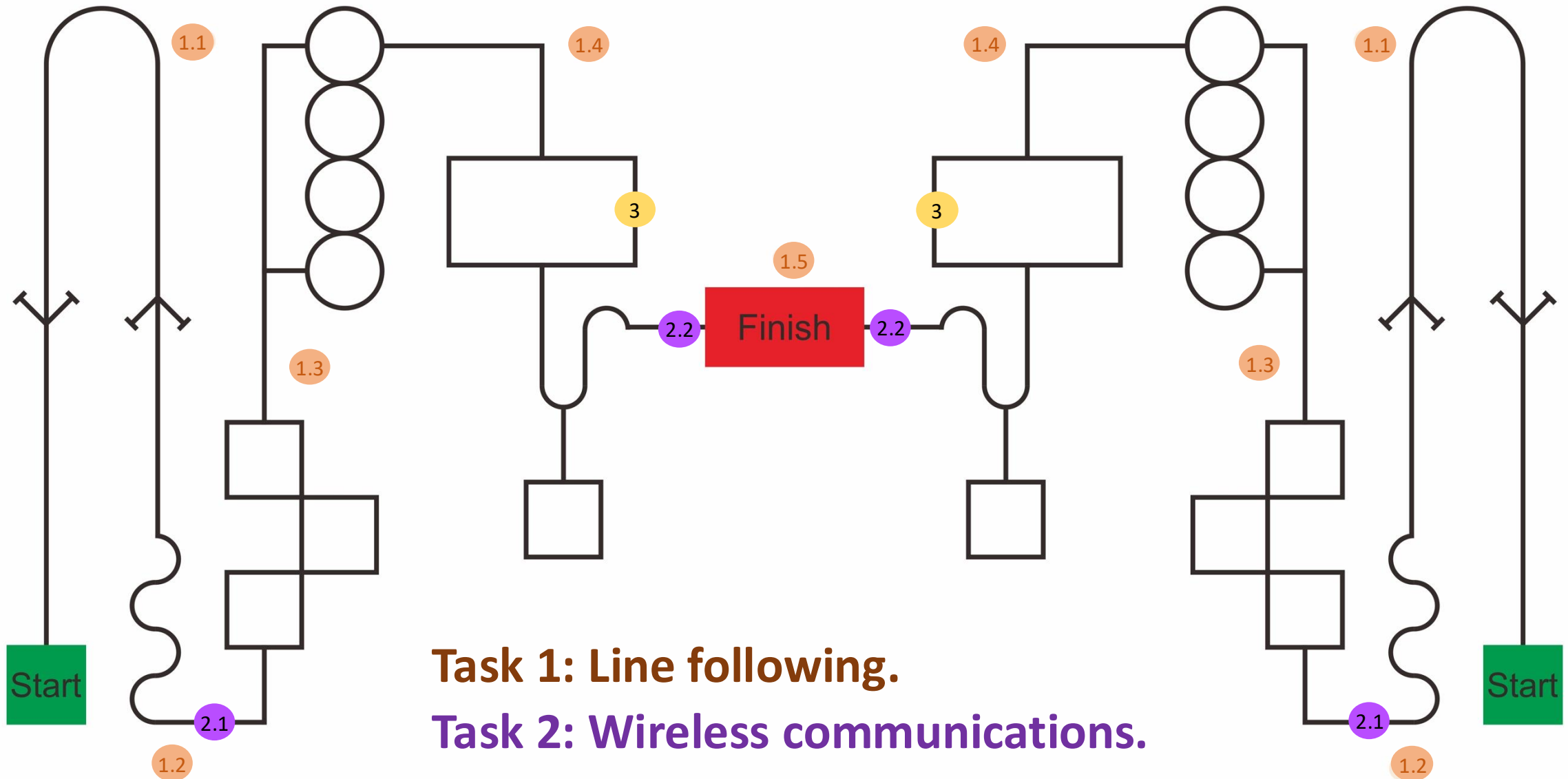
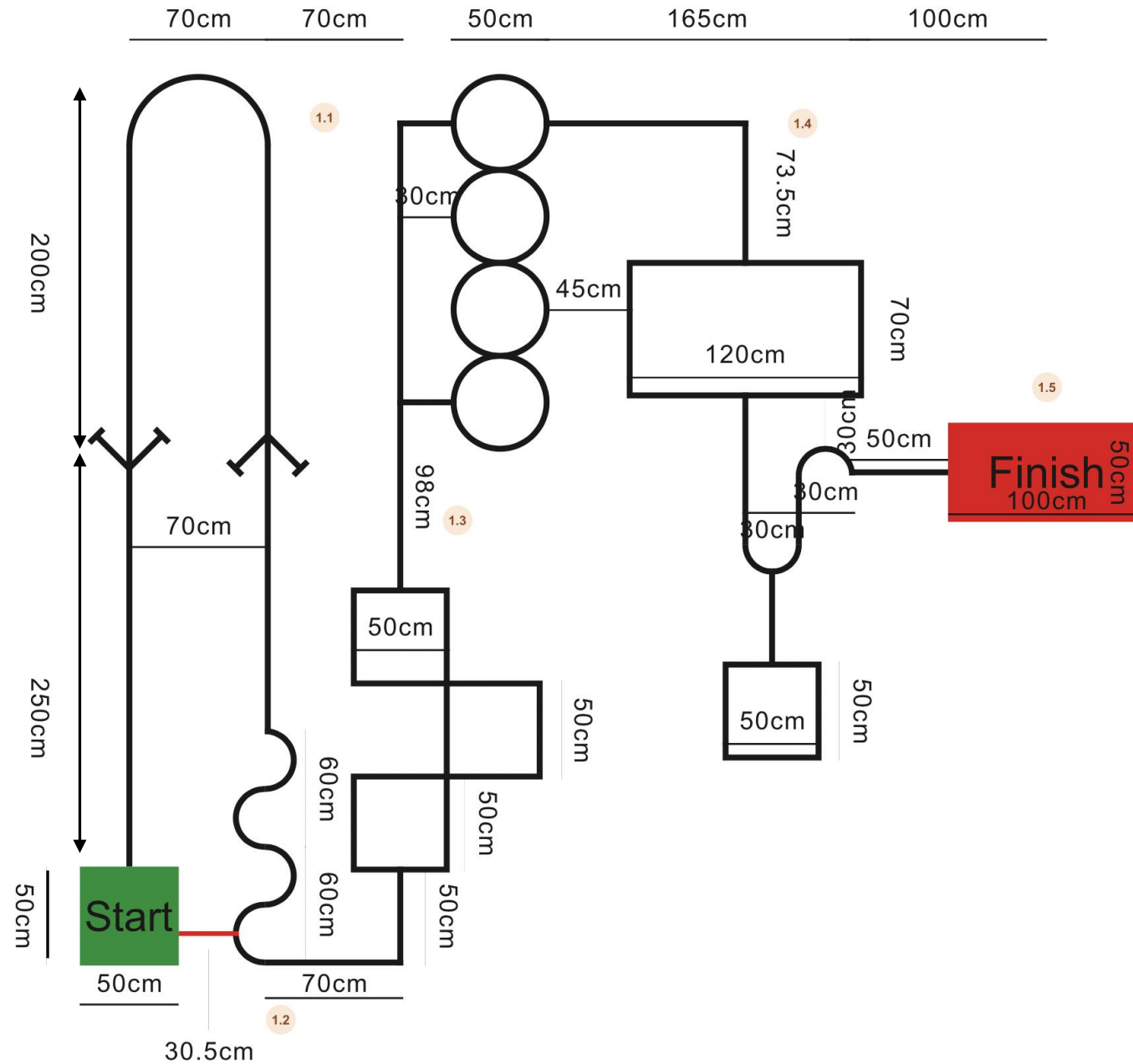


Patio - top view



Patio - top view – with dimensions

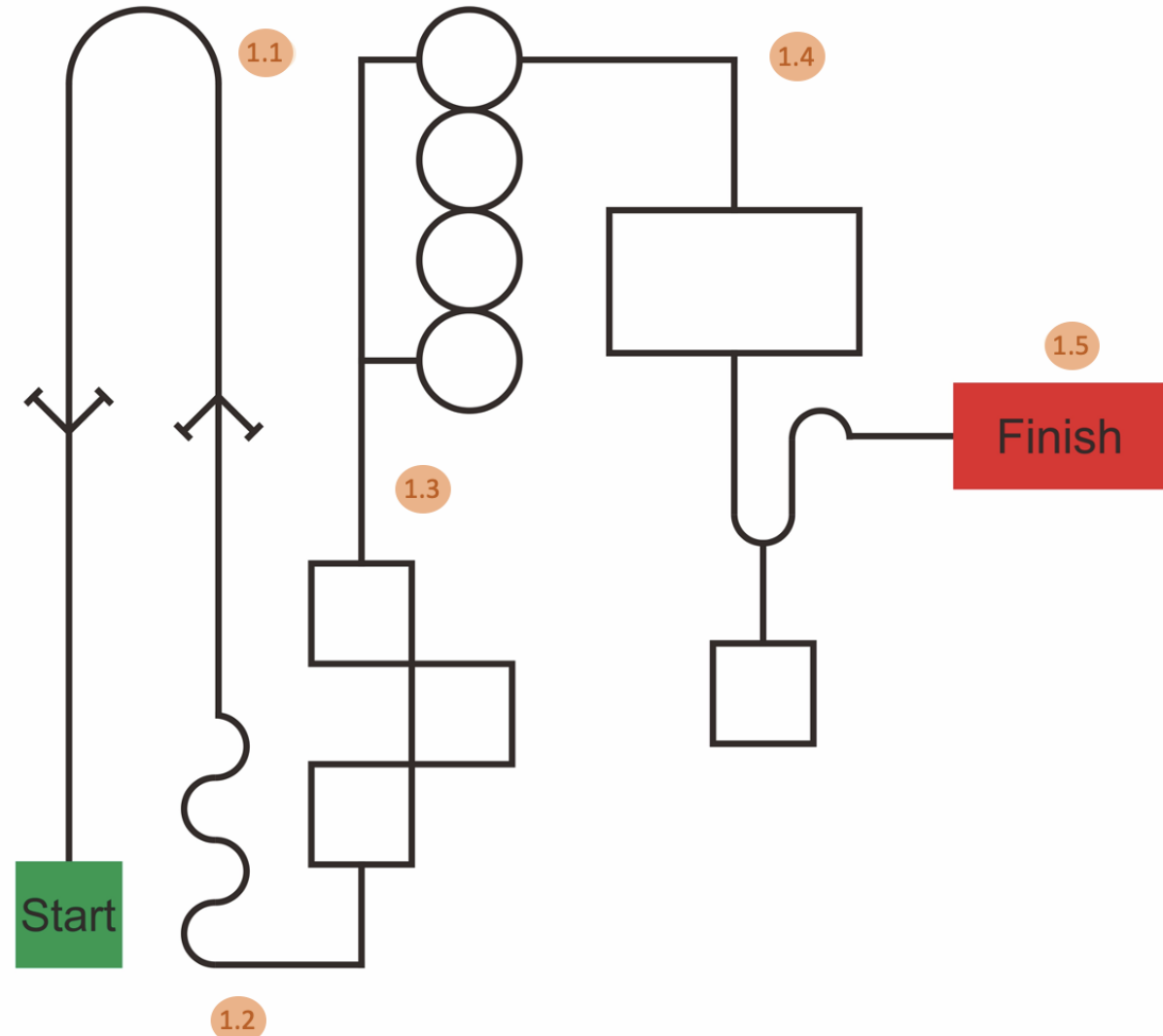


Task 1 – Line following

Task 1. From the start point (green box), your car needs to follow the black line. It also needs to navigate through the bends and boxes. Ideally, you would want to follow the shortest route to the finish box. The line will be 3 cm thick, and the green box will be 50 x 50 cm in size. This task has 5 subtasks with a combined total of 50% of the total demonstration marks.

Think!

What kind of components are needed to achieve that? A camera, IR or???

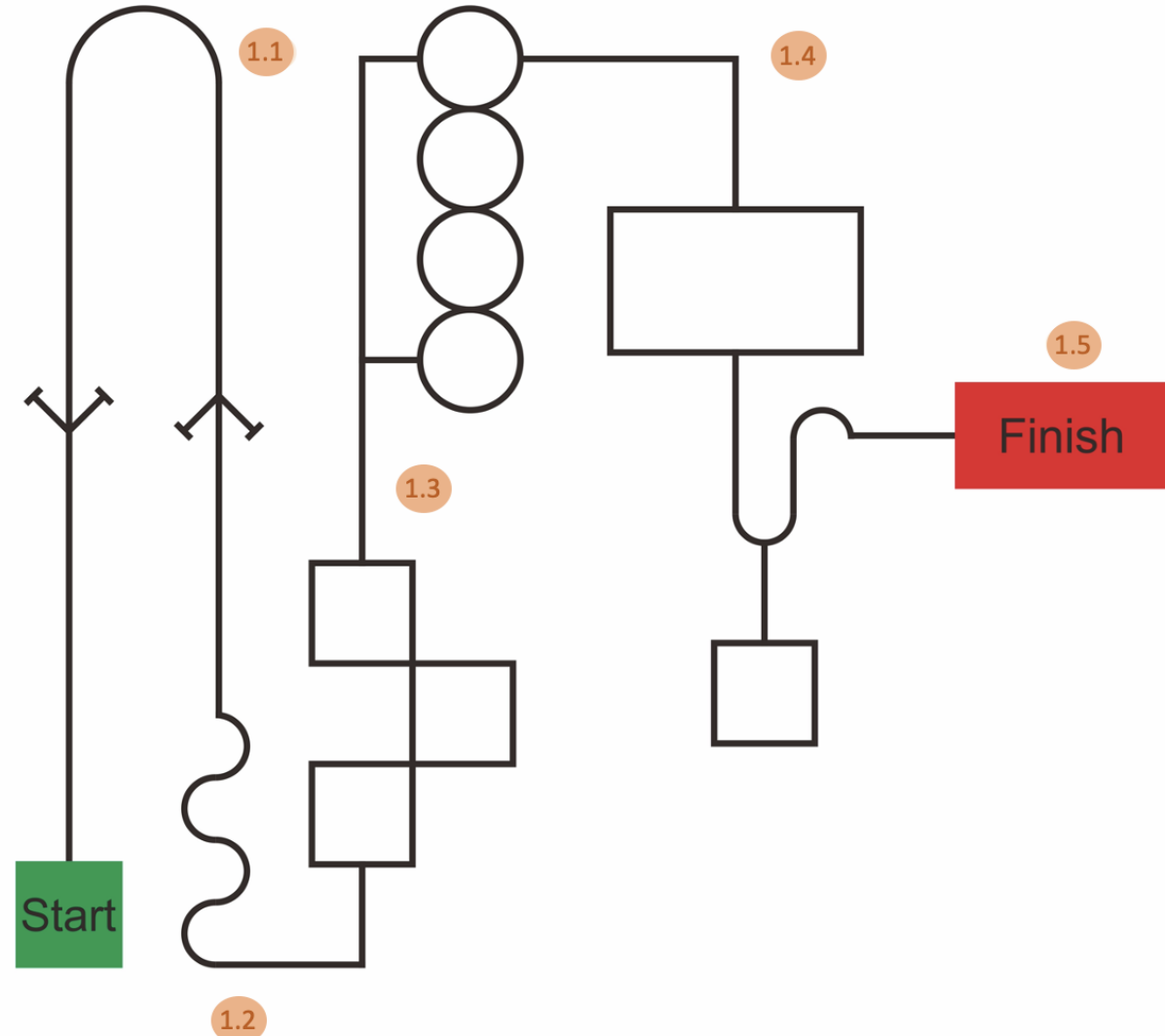


Task 1 – Line following

Task 1. The car should be on top of the line at all times. If the car fully leaves the line, the subtask must be restarted from the previous marker. e.g. if it completely leaves the line between 1.2 and 1.3, it should start again from 1.2.

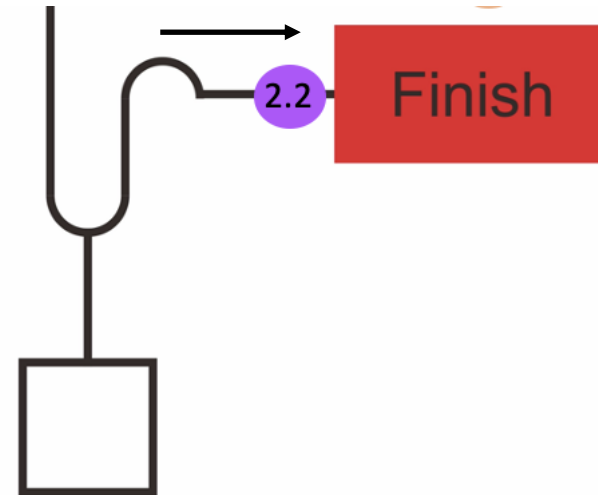
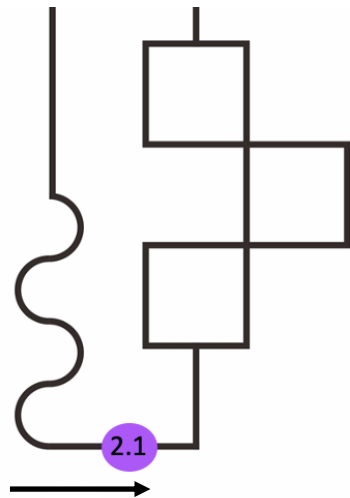
Think!

What kind of components are needed to achieve that? A camera, IR or???



Task 2 – Wireless communications

At markers 2.1 and 2.2, there will be an arch which is 50 cm high. The arch will have a wireless receiver pointing downwards. When your car passes underneath the arch, it must transmit the current time, team number and name and the time duration since the start of the race in minutes:seconds. This task will carry 20% of the total demonstration marks.



Task 2 – Wireless communications

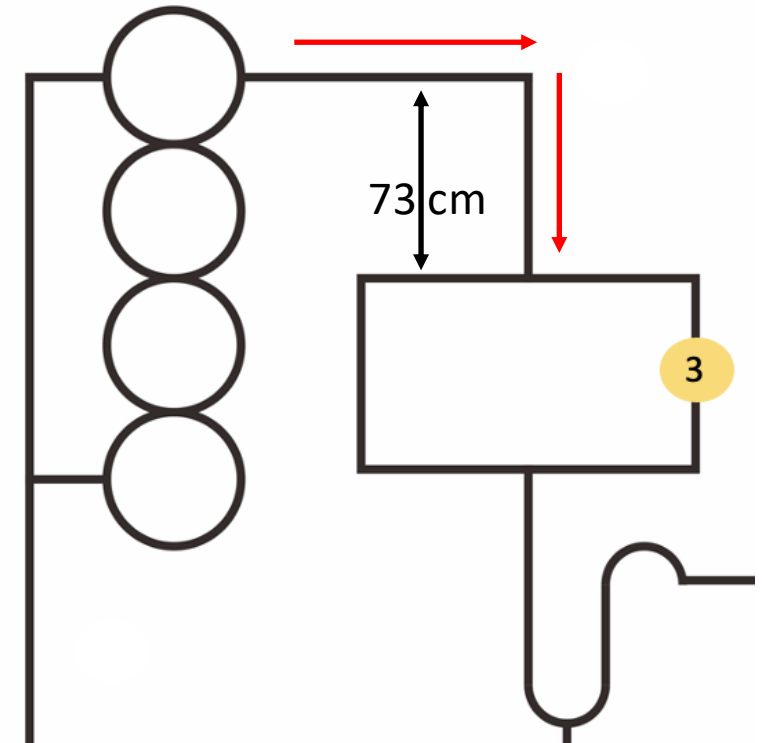
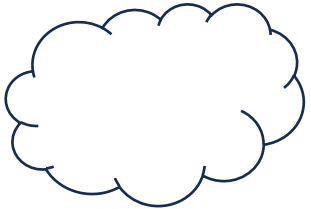
The transceiver module that you need to use is the (Ebyte EWM22A-400900BWL22S) which is based on the LoRa (Long Range) protocol.

The LoRa protocol is a low-power, long-range wireless modulation technology designed for IoT, operating in unlicensed sub-gigahertz radio bands. The centre frequency is 868 MHz and the frequency band is 850 – 930 MHz.

The receiver will be provided and integrated within the arch. You are only required to buy and program the transmitter.

Task 3 – Automotive radar

In task 3, there will be an obstacle (a metal plate which is 50 cm wide and 70 cm high) placed randomly at one side of the box, either right or left. The car should scan the box ahead and determine where the obstacle is placed before entering the box. It should use the free side of the box to pass through to the next stage. This task will carry 30% of the total demonstration marks.



For this task, a **24 GHz radar chip** must be used. Higher frequencies are also allowed but nothing below 24 GHz can be used.